Exam 2 Review

Agenda

- Covariance
- Correlation
- Discrete Probability Distributions

Which Distribution?

For each scenario, identify the most appropriate distribution: Binomial, Hypergeometric, Poisson, Exponential, Normal

1. From 100 items with 10 defectives, you select 5 without replacement and count defectives. Hypergeometric
2. A factory produces light bulbs, each with a 2% defect rate. You test 100 bulbs and count defectives. Binomial
3. Heights of adult women in a large population. Normal
4. Number of calls arriving at a call center per minute. Poisson
5. Time between consecutive calls. Exponential

Covariance and Correlation

Suppose the covariance $\sigma_{XY} > 0$.

1. What does this imply?
2. Can correlation be small?
3. Does covariance measure strength?
4. Why prefer correlation over covariance?

tend to increase together

$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$; yes

No ; it depends on units

standardized ;
interpretable ;

Compute Covariance and Correlation

Given:

$$E(X) = 10, \quad \sigma_X = 2, \quad E(Y) = 20, \quad \sigma_Y = 5, \quad E(XY) = 215$$

1. σ_{XY}

$\sigma_{XY} = E(XY) - E(X)E(Y) = 215 - 200 = 15$

2. Correlation

Coefficient: $\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} = \frac{15}{2 \cdot 5} = 1.5$

3. What is your interpretation of the correlation coefficient?

Impossible

data inconsistent

Scaling Effects

Let $Y = 3X + 7$.

1. Find σ_{XY} in terms of σ_X^2
2. Correlation ρ_{XY}
3. Why?

Independence vs Correlation

1. If independent, $\mathrm{Cov}(X, Y)$
2. Is $r = 0$?
3. Does $r = 0$ imply independence?

Discrete Distribution

$$P(X = 0) = 0.2, \quad P(X = 1) = 0.5, \quad P(X = 2) = 0.3$$

1. Expected value
2. Variance
3. Standard deviation

Correlation vs Causation

1. Does $r = 0.85$ imply causation?
2. Alternatives
3. Key idea

Binomial Model

10 fair coin tosses. Let X = number of heads.

1. Distribution
2. Mean, SD
3. $P(X \geq 7)$

Box Model

Box $\{0, 1, 2\}$, 50 draws with replacement.

1. One draw EV
2. Sum of draws $S = X_1 + \cdots + X_{50}$
3. Standard error of S